

TaxPay: Integrated Product Scope v2

Public-First AI Tax Assistant — Dual Bank Path, First-Time Filer Support & Optional Quick Filing Mode

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1 0. What Changed in This Version

This revision integrates three items from the founder’s review that were not yet reflected in the public-first refined scope:

#	Gap in Prior Version	What’s Added Here
1	Statement upload <i>replaced</i> Mono/Okra entirely	Dual bank-data path. Mono/Okra (“Instant Sync”) and statement upload now run side by side, feeding the same classifier. Users choose either or both.
2	TaxPro Max credentials were never stored, full stop	Optional Quick Filing Mode. A default “never store” guided flow remains the primary path. A clearly-labeled, opt-in, separately-consented mode allows encrypted credential storage for users who want one-tap filing, built on envelope encryption, strict access controls, and instant revocation. See Section 10 for the full security design and Section 13 for the risk trade-off this introduces.
3	No path for users without a TIN or prior FIRS/LIRS account	First-Time Filer onboarding. A guided TIN-registration-assist flow (same “we prepare it, you submit it” pattern as filing) now precedes tax calculation for users who have never registered.

Everything else in the public-first scope — statement parsing, standing orders, four-language support, voice chat — is carried forward largely as previously validated, with updates only where the three changes above touch shared architecture (data model, onboarding flow, risk register, roadmap).

2 1. Core Philosophy

The original scope narrowed TaxPay to B2B SMEs because bank integrations were fragile, FIRS has no API, and the informal economy is cash-based. That was the safe, conservative path. The real pain — the one that keeps ordinary Nigerians up at night — is personal tax chaos: not knowing what you owe, why you owe it, or how to file without paying an accountant ₦50,000 you don't have.

This build makes seven deliberate changes from the original enterprise-only plan:

#	Change	Why It Matters
1	General public first, not enterprise-only	The biggest pain is personal tax confusion, not corporate PAYE
2	Bank statement upload and live bank sync (Mono/Okra), run in parallel	Statement upload works with any bank instantly; Mono/Okra gives tech-savvy users on Tier-1 banks a one-tap “set and forget” option. Neither replaces the other.
3	Standing orders instead of auto-debit	No direct-debit API needed; user stays in control; works with every Nigerian bank
4	Semi-automated TaxPro Max filing, with an optional one-tap mode	Cuts filing from 45 minutes to 3 clicks by default; users who want zero-touch filing can opt into encrypted credential storage
5	4 languages + voice chat on WhatsApp	English, Pidgin, Hausa, Igbo — tax is scary enough in your own language
6	First-time filer support (no TIN, no prior account)	Excluding anyone without an existing FIRS/LIRS registration would exclude a large share of the informal-to-formal transition population this product is meant to serve
7	Never-store remains the default; storage is opt-in and reversible	Preserves the lower-risk posture for the majority of users while giving power users a faster path

The honest truth: this is harder to build than the B2B-only version, and harder still than the previous public-first draft, because it now carries two bank-data pipelines, a credential vault, and a registration-assist flow. But it solves a problem 15 million Nigerians actually feel, not just 25,000 HR managers, and it no longer silently excludes the unbanked-to-banked and unregistered-to-registered population in between.

3 2. Solving Step 8: The Filing Assistant

3.1 2.1 Why Step 8 Exists

Step 8: FILE MANUALLY (Employer)

- Employer downloads documents from dashboard
- Logs into TaxPro Max (or etax.lirs.gov.ng for Lagos)
- Uploads computation sheet and employee schedule
- Generates RRR for that filing
- Marks filing as submitted in TaxPay dashboard (manual confirmation)

This manual step was accepted in the B2B scope because HR managers are paid to do admin work. For individuals, this is the dealbreaker. If a regular person still has to log into TaxPro Max, navigate government websites, and upload documents manually, the product has failed them.

3.2 2.2 The TaxPro Max Reality (Confirmed)

- TaxPro Max has no public API and likely won't for years.
- Automated filing via API is technically impossible without a FIRS partnership.
- Screen-scraping violates FIRS terms and breaks whenever they update their UI.
- TaxPro Max is, however, just a web form — and web forms can be pre-filled.

3.3 2.3 Two Filing Modes

Rather than a single credential-handling policy, the filing assistant now offers two modes. **Guided Mode is the default for every user.** Quick Filing Mode is opt-in, requires separate explicit consent, and is positioned as a convenience upgrade — never a precondition for using the product.

3.3.1 Mode A — Guided Filing (Default, No Credential Storage)

1. User clicks “File My Taxes” in WhatsApp or the web dashboard.
2. TaxPay generates all required documents (computation sheet, employee schedule, payment receipt) internally.
3. The browser extension (or mobile WebView) opens TaxPro Max in a new tab and auto-fills every non-credential field: TIN, tax year/period/type, computed liability, employee schedule data.
4. The user types their own TaxPro Max password directly into TaxPro Max's own login field. TaxPay's extension never touches, reads, or transmits this keystroke — it is entered natively in the government site's own form.
5. User reviews the pre-filled data, clicks Submit, generates RRR.
6. The extension detects submission success (by watching for TaxPro Max's own confirmation page) and reports back to TaxPay.

Result: filing time drops from 45 minutes to about 3 minutes, with zero data-entry errors and zero credential exposure to TaxPay's servers.

3.3.2 Mode B — Quick Filing Mode (Opt-In, Encrypted Credential Vault)

For returning users who file every month and want a single tap:

1. During or after their first successful Guided filing, the user is offered: *“Want TaxPay to remember your TaxPro Max login for next time? Your password will be encrypted and only used to open TaxPro Max on your behalf when you tap File. You can turn this off anytime.”*

2. If the user opts in, they re-enter their TaxPro Max password once. It is immediately sealed with envelope encryption (Section 10) and never stored or logged in plaintext at any point, including in memory dumps, error logs, or backups.
3. On future filings, the user taps “File Now,” the vault decrypts the credential inside a short-lived, isolated process, injects it into the TaxPro Max login field, and discards it from memory the moment the login step completes.
4. The user still sees and confirms every pre-filled field before submission — Quick Filing Mode removes only the password re-entry step, not the review step.
5. The user can revoke stored access at any time from WhatsApp (“stop remembering my password”) or the dashboard, which triggers immediate deletion of the encrypted credential and a confirmation receipt.

This is presented as what it is: a convenience feature with a real, if well-managed, security trade-off. See Section 10 for the full encryption and access-control design, and Section 13 for how this changes the platform’s risk profile relative to a pure never-store approach.

3.3.3 The Failsafe — Filing Concierge (Revenue Model, Unchanged)

For users who don’t want to touch TaxPro Max at all, human or credential storage:

- \$1,500 per filing.
- User authorizes TaxPay as their tax agent via a digitally signed power of attorney.
- A network of verified tax practitioners files on their behalf using the pre-generated documents.
- Status tracked in WhatsApp: *“Your July tax has been filed. Reference: TAX-FILED-28472.”*

3.4 2.4 Honest Assessment

Approach	Feasibility	User Effort	Credential Exposure	Development Effort
Guided Filing (default)	8/10 — TaxPro Max forms are stable	3 minutes	None — never touches TaxPay	2–3 weeks
Quick Filing Mode (opt-in)	7/10 — adds vault, isolated-process injection, revocation flow	Under 1 minute after first setup	Encrypted at rest, decrypted only transiently	+2–3 weeks on top of Guided Filing
In-App WebView (Mobile)	7/10 — some TaxPro Max mobile quirks	5 minutes	Same as chosen mode	1–2 weeks
Filing Concierge Service	9/10 — just needs tax practitioner partners	Zero	None — human agent uses own session	1 week
Full API Integration	1/10 — requires FIRS partnership	Zero	N/A	Years of lobbying

Recommended for Hackathon MVP: Guided Filing + Filing Concierge as fallback. Quick Filing Mode is demo-able as a prototype (show the consent screen, the vault design, and a mocked decrypt-inject-discard flow) but should not go live with real user credentials until it has had a dedicated security review — this is called out explicitly in the roadmap in Section 12.

4 3. Bank Data: Dual-Path Model

4.1 3.1 Why Both Paths Coexist

The previous draft treated statement upload as a *replacement* for Mono/Okra. That undersells what each is good for. They solve different problems for different users, and running them side by side — both feeding the same transaction classifier — gives the platform its widest possible reach.

Dimension	Instant Sync (Mono/Okra)	Statement Upload
Bank coverage	Strong on Tier 1 banks, partial on Tier 2/3	100% of banks — anything that produces a PDF/CSV
Setup effort	One-time consent flow, then automatic	User exports and uploads each period
Ongoing effort	Zero — new transactions sync automatically	User re-uploads or forwards new statements each period
Trust barrier	Higher — user shares bank login via aggregator	Lower — user controls exactly what's shared
Works offline / low bank-API reliability	No — depends on aggregator uptime	Yes — works whenever the user has the file
Cost	₦5-15 per API call	Free (compute only)
Best for	Salaried users on GTBank/UBA/Access/Zenith who want zero-touch tracking	Everyone else, and anyone who doesn't want to link bank credentials to a third party

4.2 3.2 How Users Choose

At onboarding, after language selection, the user is asked how they'd like TaxPay to see their income:

"How should I track your income?"

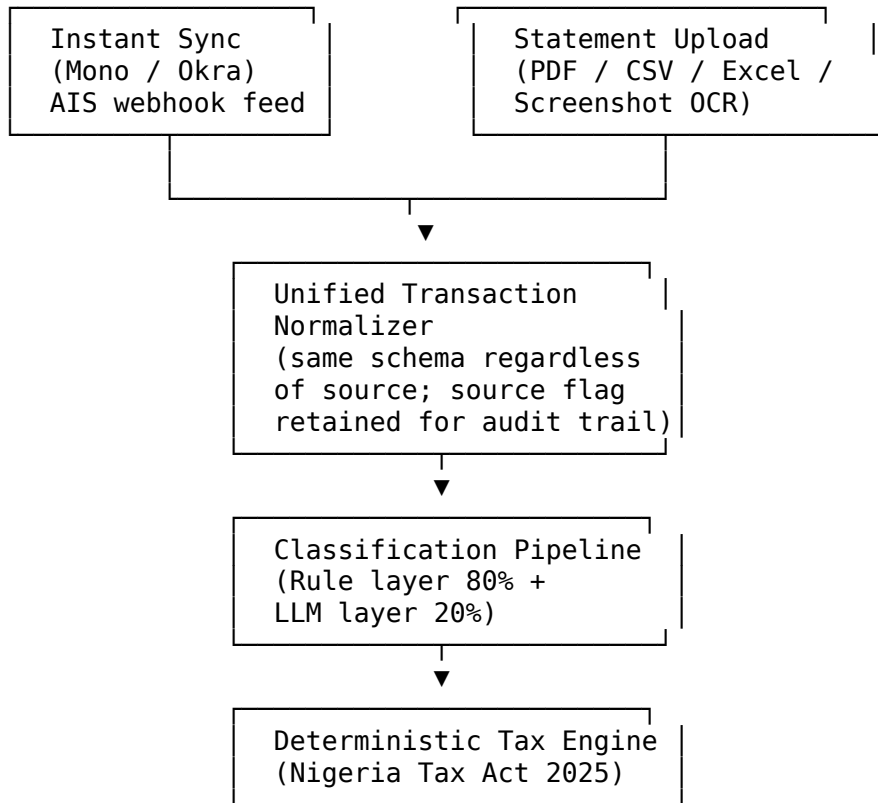
1. Instant Sync — link your bank once, I track new transactions automatically. Works best with GTBank, UBA, Access, Zenith, Kuda, Opay.
2. Upload Statements — send me your bank statement (PDF/CSV) each month or quarter. Works with every Nigerian bank.
3. Both — link what I can, and I'll ask you to upload anything Instant Sync can't reach.

Not sure? Most people start with option 2 and add Instant Sync later if they want."

Selecting "Both" is the recommended default in onboarding copy, since it maximizes coverage without forcing a trust decision either way. Users can change their selection at any time, and per-account granularity is supported: a user might link their salary account via Instant Sync but upload statements manually for a business account their bank isn't well supported by Mono/Okra.

4.3 3.3 Unified Ingestion Pipeline

Both paths converge into the same classification and tax-computation layer, so the AI, the deterministic engine, and the user-facing review step behave identically regardless of source:



4.4 3.4 Statement Upload Details (Carried Forward)

Supported formats:

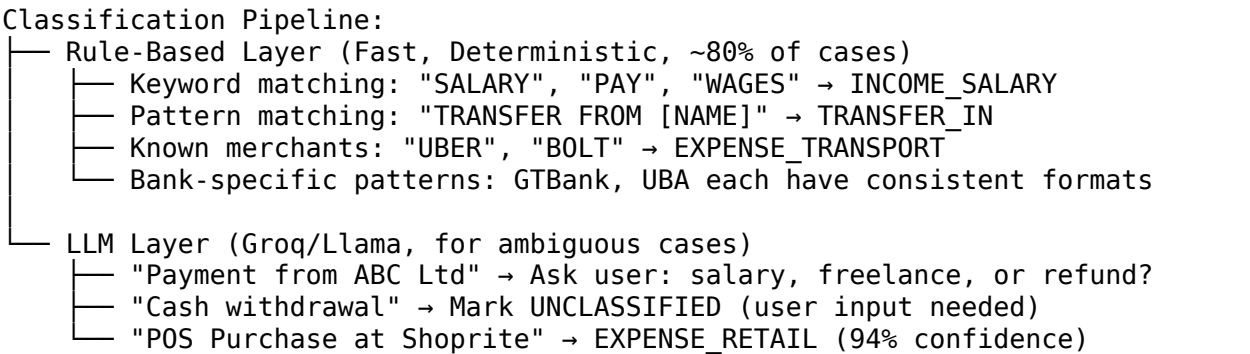
- PDF (scanned or text-based) → OCR + text extraction
- CSV (standard bank export) → Direct parsing
- Excel (.xlsx) → OpenPyXL parsing
- Screenshot → OCR fallback for users who can't export

Parsing pipeline: document detection (bank, statement type, date range) → table extraction (date, description, debit, credit, balance) → data cleaning (malformed rows, currency formatting, page breaks) → completeness validation (*"This statement is missing 15 days"*).

4.5 3.5 Instant Sync Details (Restored to Scope)

- **Provider strategy:** Mono for account information (AIS) as primary; Okra as secondary/fallback for corridors Mono doesn't cover well. Both are CBN-certified.
- **Consent flow:** standard OAuth-style bank-login consent inside Mono/Okra's own widget — TaxPay never sees or stores the bank login itself, only the resulting read-only data token.
- **Sync cadence:** daily pull of new transactions via webhook where supported, polling fallback otherwise.
- **Screen-scraping caveat:** for banks where Mono/Okra fall back to screen-scraping internally, TaxPay surfaces this to the user (*"this bank's connection is less stable — consider also uploading statements as backup"*) rather than hiding the fragility.
- **Failure handling:** if a sync fails for more than 5 days, the user gets a WhatsApp nudge offering statement upload as an immediate substitute for that period, so tax computation is never blocked on aggregator uptime.

4.6 3.6 Transaction Classification (Unchanged Logic, Now Source-Agnostic)



Category	Tax Treatment	Example
INCOME_SALARY	Taxable (PAYE)	Monthly salary from Sterling Bank
INCOME_BUSINESS	Taxable (self-employed)	Payment for website design
INCOME_FREELANCE	Taxable	Consulting fee from Client Ltd
INCOME_INVESTMENT	Taxable (dividends/interest)	Dividend from MTN shares
INCOME_RENT	Taxable	Rent received from tenant
INCOME_TRANSFER	Non-taxable	Transfer from brother
INCOME_REFUND	Non-taxable	Refund from airline
EXPENSE_BUSINESS	Deductible	Office supplies purchase
EXPENSE_MEDICAL	Potentially deductible	Hospital bill
EXPENSE_CHARITY	Deductible (if registered)	Donation to mosque/church
EXPENSE_PERSONAL	Non-deductible	Streaming subscription
TRANSFER_OUT	Non-taxable	Transfer to savings account
UNCLASSIFIED	Needs user input	Cash withdrawal

Tax calculation itself remains untouched by any of this: the LLM never calculates tax. A deterministic Python rule engine applies the Nigeria Tax Act 2025 bands and produces a full audit trail; the LLM only explains the result in natural language.

4.7 3.7 Handling the Cash Economy (Unchanged)

Bank data — synced or uploaded — cannot capture cash income. Users can self-report it, and the platform treats self-reported cash income transparently as an estimate the user can update anytime, with small-company presumptive-tax options surfaced where relevant.

5 4. First-Time Filer Onboarding (New)

5.1 4.1 The Gap

The prior draft's onboarding flow assumed every user already had a TIN and, implicitly, a Tax-Pro Max or LIRS account. That excludes a meaningful slice of the target population: young salaried employees in their first job, informal-sector operators formalizing for the first time, and freelancers who have simply never registered. Given that "first tax filing chaos" is exactly the fear this product is meant to relieve, excluding people at the very start of that journey undercuts the mission.

5.2 4.2 The Constraint

Same as filing itself: FIRS and LIRS TIN registration has no public API. Registration is a web-form process on FIRS's or LIRS's own portal. So the solution follows the same pattern already proven for filing — TaxPay prepares everything, the user submits it themselves on the government's own site.

5.3 4.3 The Flow

User: "I've never filed tax before and I don't have a TIN."

AI: "No wahala, we handle this a lot. Before we can calculate your tax, you need a Tax Identification Number (TIN) — it's free and takes about 10 minutes. I'll prepare everything and walk you through it."

To register, I need:

- Full legal name (as on your ID)
- BVN or NIN
- Phone number and email
- Home address
- Date of birth"

[User provides details via WhatsApp, one field at a time or via a short web form]

AI: "Got it. Here's your TIN registration, ready to submit:
[Pre-filled TIN registration summary]

Tap below to open the FIRS TIN registration portal — I've pre-filled everything. Just review, confirm, and submit."

[Guided-mode browser extension / WebView opens the FIRS or LIRS TIN registration form, pre-fills every field from the data collected above, user reviews and submits in their own session — same never-store-by-default credential handling as Section 2.3, Mode A]

AI: "Submitted! Your TIN is usually issued within 24–48 hours. I'll message you the moment it's active, then we can calculate your first tax bill together."

[Webhook / status-check job polls for confirmation, or user forwards the TIN confirmation email/SMS, which is parsed automatically]

AI: "Your TIN is active: [TIN]. Welcome – let's figure out what you actually owe."

[Flows directly into the standard onboarding: language, bank-data path selection, income sources, etc.]

5.4 4.4 Design Notes

- **This is a real onboarding gate, not a side quest.** A user cannot skip straight to tax calculation without either an existing TIN or completing this flow — the deterministic tax engine needs a TIN to eventually file, even though it isn't needed to *estimate* liability. TaxPay can and does show an estimate before TIN issuance ("here's roughly what you'd owe once registered") so the user gets value immediately, clearly labeled as provisional.
- **State-level nuance:** Lagos residents register via LIRS's e-tax portal; everyone else registers federally via FIRS. The assistant detects state from the user's address and routes to the correct portal, mirroring the multi-state handling already built for filing.
- **No TIN yet, income already flowing:** for users mid-registration who need to start budgeting immediately, the standing-order and reminder features (Section 5) work off the provisional estimate and simply get relinked to the real liability once the TIN is active and the first real computation runs.
- **Concierge fallback exists here too:** users who don't want to touch the FIRS/LIRS registration form themselves can route TIN registration through the same Filing Concierge network at a small flat fee, consistent with the filing concierge model.

6 5. Standing Orders (Carried Forward, Unchanged)

6.1 5.1 Why Auto-Debit Was Rejected

Nomba's Direct Debit API "pulls funds from customer accounts with prior authorization." For public users this creates three problems: deep distrust of anything that can pull money from an account without a per-transaction confirmation; incomplete bank coverage; and a bad failure mode where a failed pull results in a late-payment penalty from LIRS with no warning.

6.2 5.2 Standing Orders: User-Controlled, Bank-Agnostic

A standing order is an instruction the user gives *their own bank* to transfer a fixed amount to a specified account on a specified date, every month. It is user-controlled, works with every Nigerian bank including microfinance banks, revocable directly through the user's bank with no need to contact TaxPay, and predictable for reconciliation.

Step 1: TaxPay calculates monthly tax

→ "Your estimated PAYE for July 2026 is ₦47,500"

Step 2: Generate standing order instruction

→ Payee: "TaxPay Tax Collection"

→ Account: [Nomba Virtual Account for this user/period]

→ Amount: ₦47,500 | Frequency: Monthly | Start: Aug 10, 2026

→ Reference: "TAXPAY-[UserID]-PAYE"

Step 3: User sets it up at their bank branch, in their banking

app, or via USSD (e.g. *737*50*Amount*AccountNumber*

Frequency# for GTBank)

Step 4: User confirms setup (screenshot), TaxPay marks ACTIVE

Step 5: Nomba webhook confirms each month's payment automatically;

TaxPay sends a WhatsApp confirmation and updates the compliance dashboard

Variable-income users (freelancers, business owners) get a recommended standing-order amount based on a trailing 3-month average, with quarterly reconciliation that adjusts the amount up or down and settles any overpayment or underpayment at year-end.

7 6. Multi-Language Architecture & Voice Chat (Carried Forward, Condensed)

7.1 6.1 Why Four Languages Matter

Language	Speakers (approx.)	Tax Complexity Tolerance
English	60 million	Medium — formal terms are confusing
Pidgin	75 million	Highest — everyday language
Hausa	50 million	Low for English tax terms
Igbo	30 million	Low for English tax terms

Pidgin is the killer feature. A tax bot that says *“How much you dey owe this month?”* instead of *“What is your tax liability for the current fiscal period?”* wins instantly with the audience this product is built for.

7.2 6.2 Three-Tier Response Architecture

1. TAX CALCULATION (Deterministic – identical for all languages)
 - └ Python rule engine → structured result JSON
2. EXPLANATION TEMPLATES (Per-language, pre-written)
 - └ Covers ~80% of common queries
3. LLM GENERATION (For complex/conversational responses)
 - └ Groq Llama 4 with language-specific system prompts

Language is detected via FastText (offline, sub-millisecond) with keyword fallback, and the user can override per-message with /english, /pidgin, /hausa, /igbo.

7.3 6.3 Voice Chat

WhatsApp voice notes are transcribed via Whisper API (STT) and responses synthesized via ElevenLabs (TTS), with per-language voices. Text fallback is always available (“I no hear you well. Abeg type am or send am again.”). Voice is priced as a premium add-on (~\$500/month) since STT/TTS cost scales with usage; at 10,000 users, total voice infrastructure cost is roughly \$1,600/month, manageable as an opt-in tier rather than a default.

8 7. Complete User Journeys

8.1 7.1 Amaka — Salaried Employee, Instant Sync

Amaka, 28, works at a bank in Lagos, earns ₦600,000/month, already has a TIN.

- Onboarding: selects English, chooses "Instant Sync" for bank data, links GTBank via Mono in under a minute
- AI confirms detected salary of ₦600,000/month
- Rent (₦1,200,000/yr) and pension (8%) captured
- Monthly PAYE calculated: ₦47,500
- Standing order set up
- Day 10: transfers ₦47,500; Nomba webhook confirms
- Day 11: "Want me help you file? Takes 2 minutes."
 - Guided Filing Mode: extension pre-fills TaxPro Max, Amaka types her own password into TaxPro Max directly, reviews, submits, generates RRR
- After her third filing, TaxPay offers Quick Filing Mode; Amaka opts in for future months and files with one tap
- Year-end: Form H1 auto-prepared, same 2-minute filing flow

8.2 7.2 Chinedu — Freelancer, Statement Upload Only

Chinedu, 35, web developer in Enugu, earns variable income, already has a TIN, prefers not to link bank credentials to any third party.

- Onboarding: selects "Upload Statements" only
- Uploads UBA + Payoneer statements (last 6 months)
- AI classifies client payments, transfers, equipment purchases; freelance income averages ₦520,000/month, ₦85,000/month in deductible equipment/internet costs
- Standing order recommended at ₦65,000/month
- Quarterly reconciliation adjusts the standing order as income fluctuates
- Continues filing via Guided Mode each period; never enables Quick Filing Mode

8.3 7.3 Ibrahim — First-Time Filer, No TIN (New)

Ibrahim, 24, just started his first job in Kano as a junior accountant, earns ₦280,000/month, has never registered for tax.

- Onboarding: selects Hausa
- "Do you have a TIN?" → No
- First-Time Filer flow: TaxPay collects name, BVN, phone, email, address, DOB via WhatsApp
- TaxPay pre-fills the FIRS TIN registration form; Ibrahim opens it via Guided Mode, reviews, submits in his own session
- While waiting for TIN issuance, TaxPay shows a provisional estimate: "Once your TIN is active, your PAYE should be around ₦8,200/month based on what you've told me – this is an estimate, not your final figure."
- 36 hours later: TIN confirmation SMS forwarded to TaxPay, parsed automatically; TIN activated in Ibrahim's profile

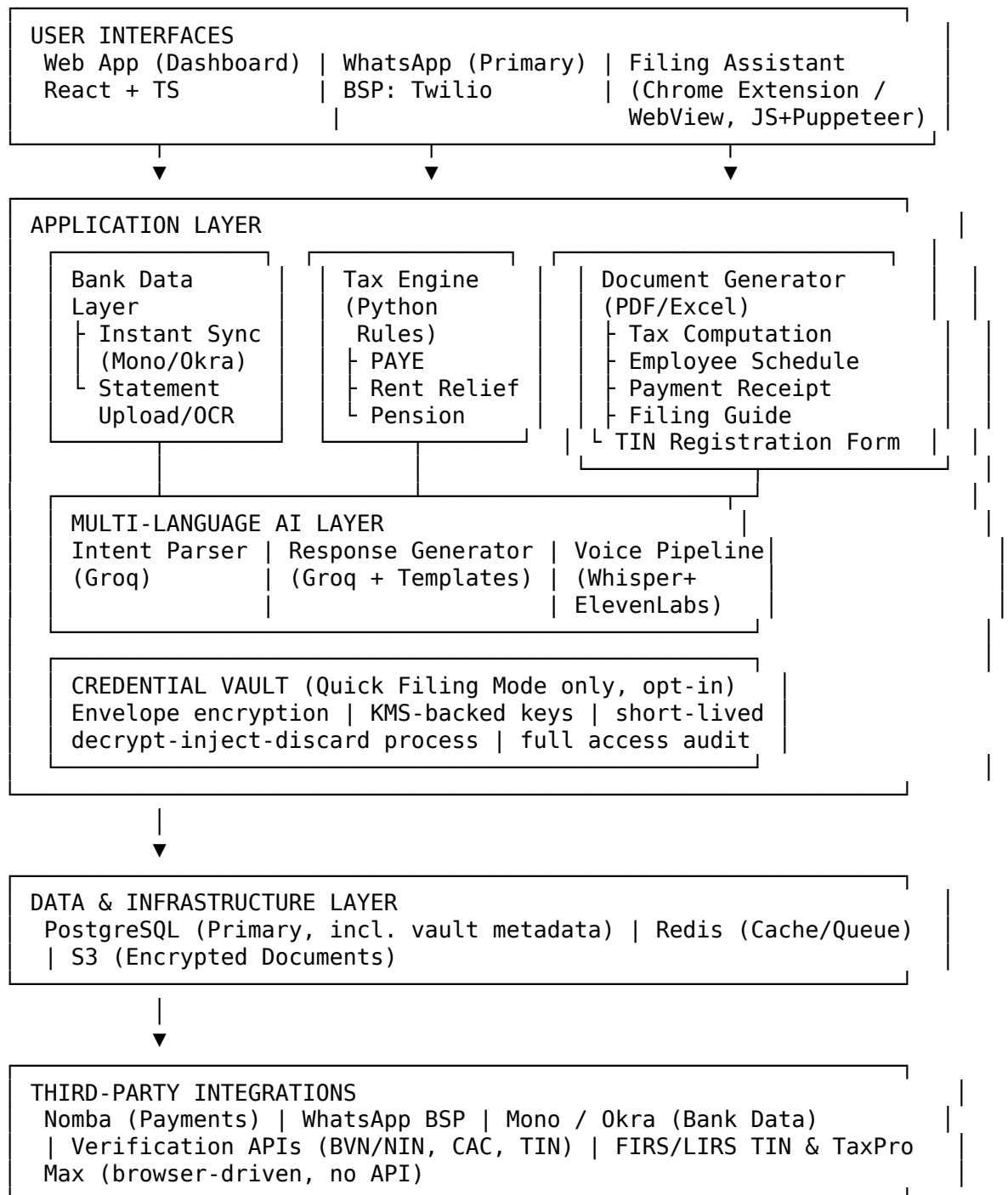
- Standard onboarding resumes: bank-data path selection (Ibrahim chooses "Both" – links his Access Bank salary account via Instant Sync, and uploads a statement for a side hustle account at a smaller bank Mono doesn't support)
- First real PAYE calculation runs, standing order set up
- First filing at month-end via Guided Mode

8.4 7.4 Hajiya Amina — Small Business Owner, Mixed Cash and Transfer

Unchanged from the prior draft: Hausa-language onboarding, CAC/TIN verification, employee PAYE plus mixed cash/transfer income handling, bank statement upload for the transfer side, self-reported estimates for the cash side.

9 8. Revised Technical Architecture

9.1 8.1 High-Level System Diagram



9.2 8.2 Technology Stack (Updated)

Layer	Technology	Why
Frontend (Web)	React 18 + TypeScript + Tailwind	Standard, fast, mobile-responsive
Frontend (Extension)	Vanilla JS + Puppeteer (Manifest V3)	Lightweight, injects into TaxPro Max / FIRS TIN portal
Backend API	FastAPI (Python) + async SQLAlchemy	Native async, strong for webhook handling
Tax Engine	Pure Python (no LLM) + Pandas	Deterministic, auditable, testable
Statement Parser	pdfplumber + PyMuPDF + Tesseract OCR	Handles Nigerian bank statement formats
Bank Sync	Mono (primary AIS) + Okra (fallback)	Instant Sync path, CBN-certified
Database	PostgreSQL 16 + JSONB	ACID compliance, flexible tax-form schema
Cache/Queue	Redis (Streams + Sorted Sets)	Sessions, standing-order reminders, vault access throttling
Document Storage	AWS S3 + Server-Side Encryption	Encrypted statements, generated documents
Credential Vault	AWS KMS envelope encryption + isolated decrypt worker	Separates key management from data storage; short-lived plaintext exposure only
LLM (Chat)	Groq (Llama 4 Scout)	Low latency, cheap
LLM (Complex)	GPT-4.1 Mini (fallback)	Better JSON reliability
STT / TTS	OpenAI Whisper / ElevenLabs	Nigerian-accent handling, natural voices
Language Detection	FastText (pre-trained)	Sub-ms, offline
WhatsApp BSP	Twilio or 360Dialog	Reliable, good Nigerian support
Payments	Nomba (Virtual Accounts + Webhooks)	Works today, no FIRS integration needed
Hosting	AWS (ECS Fargate) + CloudFront	Scalable, Nigerian edge locations
Auth	Clerk or Supabase Auth	BVN verification via custom webhook
Monitoring	Sentry + Datadog	Error tracking, performance monitoring, vault access alerting

9.3 8.3 API Endpoints (Key, Updated)

Authentication

POST /api/v1/auth/register	# BVN + phone registration
POST /api/v1/auth/verify-otp	# OTP verification
POST /api/v1/auth/set-language	# Preferred language

First-Time Filer / TIN Registration (NEW)

POST /api/v1/tin/prepare	# Collect + pre-fill TIN registration
POST /api/v1/tin/assistant-launch	# Launch guided TIN registration
POST /api/v1/tin/confirm	# Confirm TIN issued (webhook or parsed SMS)
GET /api/v1/tin/status	# Check registration status

```

# Bank Data – Dual Path (UPDATED)
POST /api/v1/bank/link-instant-sync      # Initiate Mono/Okra consent flow
GET  /api/v1/bank/sync-status            # Check Instant Sync health per account
POST /api/v1/statements/upload           # Upload bank statement
GET  /api/v1/statements/:id/status       # Processing status
GET  /api/v1/transactions                # Unified transactions (any source)
PUT  /api/v1/transactions/:id/classify   # User correction

# Tax Calculation
POST /api/v1/tax/calculate                # Calculate tax from linked data
GET  /api/v1/tax/liability               # Current period liability
GET  /api/v1/tax/breakdown               # Detailed breakdown
GET  /api/v1/tax/history                  # Historical calculations

# Standing Orders
POST /api/v1/standing-orders/create
GET  /api/v1/standing-orders/template
PUT  /api/v1/standing-orders/confirm

# Payments (Nomba)
POST /api/v1/payments/virtual-account
GET  /api/v1/payments/status
POST /api/v1/payments/webhook

# Filing
POST /api/v1/filing/prepare
POST /api/v1/filing/assistant-launch     # Mode A or Mode B, per user setting
POST /api/v1/filing/confirm
GET  /api/v1/filing/documents

# Credential Vault (Quick Filing Mode only, NEW)
POST /api/v1/vault/enroll                 # Opt-in, capture + seal credential
POST /api/v1/vault/use                    # Decrypt-inject-discard for one filing
DELETE /api/v1/vault/revoke               # Immediate deletion, confirmation receipt
GET  /api/v1/vault/access-log             # User-facing log of every vault use

# Compliance
GET /api/v1/compliance/status
GET /api/v1/compliance/reminders
GET /api/v1/compliance/certificate

# Voice
POST /api/v1/voice/process
POST /api/v1/voice/synthesize

# WhatsApp Webhooks
POST /api/v1/whatsapp/webhook
POST /api/v1/whatsapp/status

```

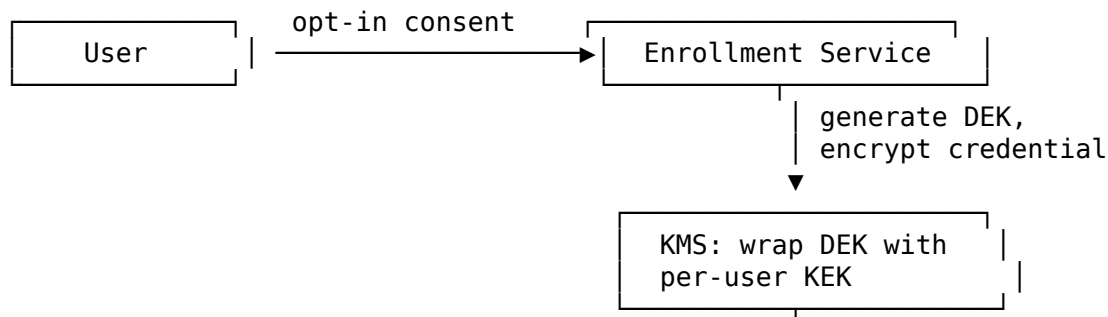
10 9. Security & Credential Vault Design (New)

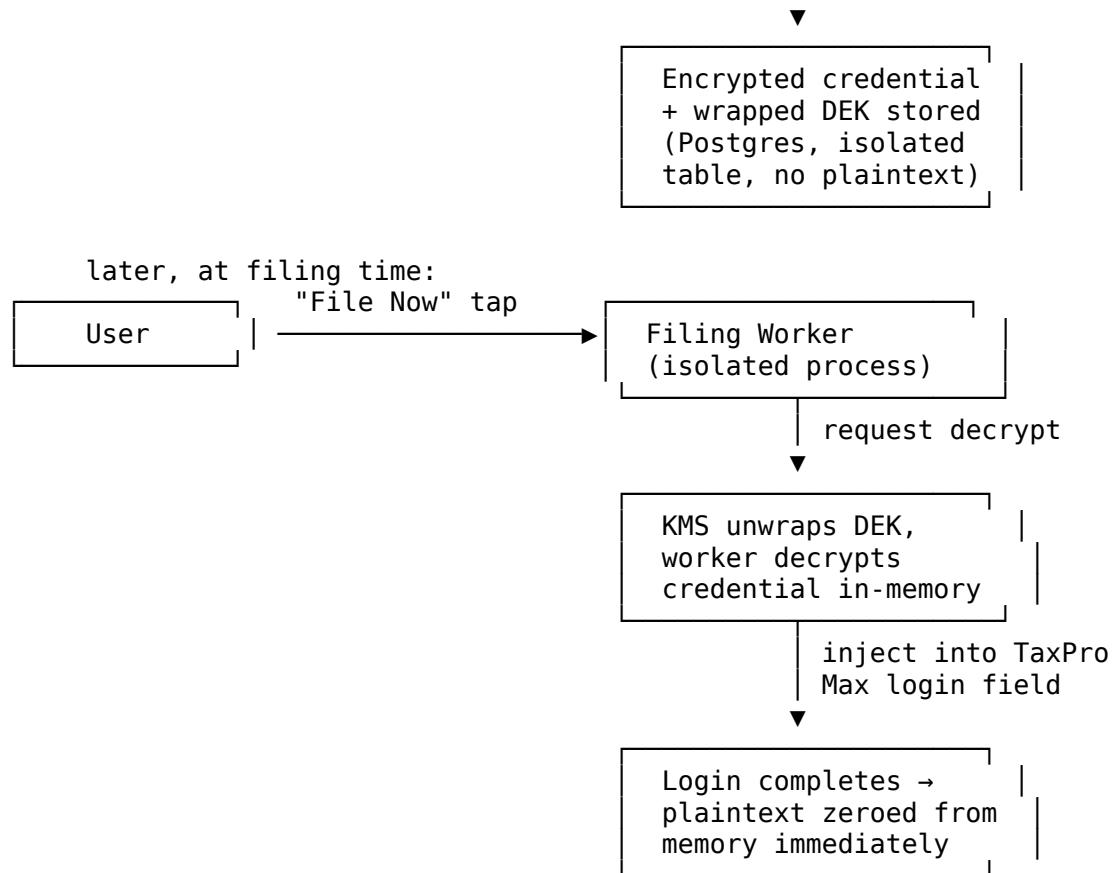
This section exists because Quick Filing Mode changes the platform’s risk posture, and that trade-off deserves to be designed deliberately rather than bolted on. **Guided Mode (Section 2.3, Mode A) remains the default for every user and requires none of this — it never touches a TaxPro Max credential at all.** Everything below applies only to the opt-in Quick Filing path.

10.1 9.1 Design Principles

1. **Opt-in, not default.** No user is ever asked for their TaxPro Max password until after they’ve completed at least one Guided filing and explicitly chosen to enable Quick Filing Mode.
2. **Separate, specific consent.** The consent screen names exactly what is stored, why, how it’s protected, and how to remove it — not bundled into general terms of service.
3. **Envelope encryption, not a shared secret.** The credential is encrypted client-side (or at the edge, before it ever reaches application logic) with a per-user Data Encryption Key (DEK), and the DEK itself is encrypted with a Key Encryption Key (KEK) held in a managed KMS (AWS KMS or equivalent HSM-backed service). The application layer never has direct access to raw key material.
4. **Short-lived plaintext exposure.** Decryption happens only inside an isolated, single-purpose worker process at the moment of filing. The plaintext credential exists in memory for the duration of the TaxPro Max login step only, is never written to disk or logs, and is explicitly zeroed from memory afterward.
5. **No plaintext anywhere, ever.** Not in application logs, not in error traces, not in database backups, not in analytics events. Structured logging is configured with field-level redaction for any parameter that could carry a credential.
6. **Least privilege access.** Only the filing-assistant worker service has permission to invoke the vault’s decrypt operation, and only for the specific user session that requested a filing. No human operator, support agent, or admin console has a path to plaintext credentials — support tooling can trigger *revocation* but never *retrieval*.
7. **Instant, user-controlled revocation.** “Stop remembering my password,” said in WhatsApp or clicked in the dashboard, deletes the encrypted credential and its DEK immediately, with a confirmation receipt and timestamp.
8. **Full audit trail.** Every vault enrollment, use, and revocation is logged with actor, timestamp, and outcome, and the user can see their own access log at any time via `/api/v1/vault/access-log`.
9. **Independent of everything else.** A breach of the main application database does not expose vault contents, because DEKs are individually wrapped by a KEK the application layer cannot read directly — compromising the app doesn’t compromise KMS.

10.2 9.2 Flow Diagram





10.3 9.3 What This Does Not Solve

This design reduces exposure; it does not eliminate the underlying trade-off. TaxPay becomes a system that, for opted-in users, is capable of authenticating to a government tax portal on their behalf. That is meaningfully different from a system that never can, regardless of how well the vault is engineered. Before Quick Filing Mode goes live with real credentials (as opposed to being demoed as a prototype), it should get:

- **Independent security review** of the vault implementation specifically, not just the wider platform.
- **Legal review** of whether encrypted storage of TaxPro Max credentials is consistent with FIRS's terms of service, since "never automate filing via credential reuse" concerns exist independent of encryption quality.
- **NDPR classification review**, since credential storage likely elevates the platform's Data Protection Compliance Organization (DPCO) audit scope.

This is reflected in the roadmap (Section 12) by keeping Quick Filing Mode out of the hackathon MVP and out of the V1 default path.

11 10. Revenue Model (Public-First, Updated)

11.1 10.1 Freemium Structure

Tier	Price	Target	What's Included
Free	₦0	Everyone	Tax calculator, 5 WhatsApp queries/month, basic tax education, TIN registration assist
Basic	₦2,000/month	Salaried employees	Unlimited queries, 1 bank statement or Instant Sync account, Guided filing
Pro	₦5,000/month	Freelancers, multiple income	4 statements/month or multi-account sync, business tax calculation, standing order setup, multi-state filing
Business	₦15,000/month	SMEs with employees	Up to 10 employees PAYE, compliance dashboard, accountant access
Enterprise	₦50,000+/month	Larger businesses	Unlimited employees, API access, dedicated support, SLA
Filing Concierge	₦1,500/filing	Anyone hands-off	Human agent files on user's behalf
Quick Filing Add-On	₦500/month	Repeat filers who opt in	One-tap filing via credential vault, once released beyond prototype

Instant Sync and statement upload are both included at every paid tier — neither is gated behind a higher tier than the other, since forcing a trust decision in order to unlock functionality would undermine the point of offering both.

11.2 10.2 Why This Pricing Works

Free tier drives adoption with no barrier to try. ₦2,000/month is less than one lunch in Lagos. ₦5,000/month is still cheaper than one accountant visit. Business and Enterprise tiers preserve the B2B revenue stream from the original scope. Filing Concierge and the Quick Filing add-on both turn what could be viewed as limitations (manual filing friction, credential-storage risk) into either a service or a clearly-priced convenience upgrade.

11.3 10.3 Revenue Projections (Conservative, Unchanged Trajectory)

Metric	Month 6	Month 12	Month 18	Month 24
Free users	5,000	25,000	60,000	120,000
Basic subscribers	200	1,500	4,000	8,000
Pro subscribers	50	400	1,200	3,000
Business accounts	20	150	500	1,200
Filing concierge/mo	100	800	2,500	6,000
Monthly Revenue	¥1.1M (~\$670)	¥11.8M (~\$7,100)	¥41M (~\$24,600)	¥96M (~\$57,600)
Annual Run Rate	\$8,040	\$85,200	\$295,200	\$691,200

Key assumption: 5-8% free-to-paid conversion, in line with freemium fintech norms. Quick Filing Mode revenue is deliberately excluded from these projections until it has cleared the security and legal review in Section 9.3.

12 11. Implementation Roadmap (Updated)

12.1 11.1 Hackathon MVP (48-72 Hours) — What's Demo-Able

Goal: working end-to-end flow from bank data (either path) → tax calculation → Guided filing assistant launch, plus a first-time-filer demo and a Quick Filing Mode *prototype* (UI and vault architecture shown, not connected to a real TaxPro Max session).

Feature	Hours	Priority
WhatsApp bot (basic) with English + Pidgin	6	Must have
PDF/CSV bank statement parser (2-3 banks)	8	Must have
Mono sandbox Instant Sync (1 bank)	4	Must have
Transaction classifier (rule-based)	4	Must have
Deterministic PAYE engine (Lagos rates)	4	Must have
Tax explanation in Pidgin + English	3	Must have
Nomba virtual account creation	2	Must have
Payment webhook handler	2	Must have
Guided Filing Chrome extension (basic auto-fill)	6	Must have
First-Time Filer / TIN-prepare flow (mocked submission)	4	Must have
Document generator (PDF receipt)	3	Nice to have
Quick Filing Mode UI + vault architecture demo (no live credentials)	4	Nice to have
Voice note STT (basic)	4	Nice to have
Hausa + Igbo templates	2	Stretch

Total: ~54 hours — achievable in a 3-day hackathon with 4 developers, with Quick Filing Mode explicitly scoped as a design/demo item rather than a live feature.

12.2 11.2 V1 (Month 1-3): Production-Ready

- All 4 languages fully templated (200+ queries each); voice chat polished.
- Statement parser supports 15+ Nigerian banks; Instant Sync live for Mono + Okra across Tier 1 banks.
- First-Time Filer flow live end-to-end, including LIRS routing for Lagos residents.
- Multi-state tax engine (Lagos, Rivers, Kano, FCT, Delta).
- Guided Filing Assistant handles the full TaxPro Max flow.
- Filing Concierge service live with 5 tax practitioner partners.
- Standing order generation + reminder system.
- NDPR compliance basics; DPCO engagement started given the credential-adjacent roadmap ahead.

12.3 11.3 V2 (Month 3-6): Scale

- AI-powered transaction classification refinements across both bank-data paths.
- Cash income reporting for the informal sector.
- Annual Form H1 auto-generation.
- Compliance dashboard.
- **Quick Filing Mode security and legal review completed** (Section 9.3); if cleared, limited beta release to opted-in Basic/Pro subscribers with the vault design in Section 9.
- Mobile filing assistant (in-app WebView).

12.4 11.4 V3 (Month 6-12): Platform

- All 36 states.
- VAT calculation and filing.
- Withholding tax management.
- API access for banks and payroll providers.
- White-label for accounting firms.
- FIRS partnership pursuit.
- Expand to Ghana/Kenya.

13 12. Updated Risk Assessment

Risk	Probability	Impact	Mitigation
TaxPro Max changes UI, breaking filing assistant	7/10	8/10	Monitor weekly; graceful fallback to manual guide; Filing Concierge as backup
Bank statement formats change	4/10	5/10	Parser is modular; add new formats as needed
Mono/Okra coverage gaps or screen-scraping fragility	5/10	5/10	Statement upload always available as immediate substitute; surface connection stability to users
Credential vault breach (Quick Filing Mode)	2/10	9/10	Envelope encryption, KMS-backed keys, isolated decrypt-inject-discard worker, no plaintext logging, independent security review before live release, opt-in only so blast radius is limited to enrolled users
FIRS terms-of-service exposure from credential reuse	3/10	7/10	Legal review before Quick Filing Mode leaves prototype stage; Guided Mode (no credential storage) remains the default and primary path regardless
TIN registration data mishandled (BVN, NIN, address collected pre-TIN)	3/10	7/10	Same encryption-at-rest and access controls as other PII; data minimization — collect only what the TIN form requires; short retention if registration is abandoned
STT accuracy poor for Hausa/Igbo	6/10	5/10	Text fallback always available; improve models over time

Risk	Probability	Impact	Mitigation
Users don't trust statement upload or Instant Sync	5/10	7/10	Offer both paths so neither trust barrier blocks onboarding; explain data handling; NDPR compliance
Standing order adoption low	4/10	6/10	Push manual payment as fallback; integrate USSD for major banks
Freemium conversion too low	5/10	7/10	Aggressive value demonstration; Filing Concierge as entry-point paid feature
FIRS/LIRS blocks filing or TIN-registration assistant	3/10	9/10	Frame as an accessibility tool, not automation; pursue formal partnership; Filing/Registration Concierge fallback
NDPR compliance complexity	4/10	6/10	Engage DPCO early; privacy by design; data minimization
WhatsApp BSP rejection	2/10	8/10	Use 360Dialog or a local BSP as alternative; web app as fallback
Voice chat cost too high	5/10	4/10	Premium feature pricing; text-first with voice upgrade

The two risks specific to this revision — credential vault breach and FIRS terms-of-service exposure — are rated lower probability than most items on this list precisely because Quick Filing Mode ships opt-in, behind a security and legal review, with Guided Mode as the unconditional default. That structure is what keeps their probability low; removing it would raise both.

14 13. What This Build Actually Delivers

For the Nigerian who earns ₦400,000/month and has never filed tax, and doesn't even have a TIN:

"Before TaxPay: I don't know what tax is, why I should pay it, or how to file. I don't even know where to start getting registered. Every January I worry. An accountant wants ₦30,000 just to talk to me.

With TaxPay: I told the bot in Pidgin I've never registered. It walked me through getting my TIN in ten minutes. Once that was active, I linked my bank and it told me exactly what I owe and why. It set up a standing order. It filed for me on TaxPro Max in 3 clicks. I get a receipt and a compliance badge. I sleep well."

For the freelancer who doesn't want to link bank credentials to anyone:

"With TaxPay: I upload my bank statements myself, every month. Nobody has my banking password. It classifies everything, tells me my actual tax liability, sets up a standing order, and at year-end it reconciles and files. When I file, I type my own TaxPro Max password into TaxPro Max's own page — TaxPay never sees it."

For the repeat filer who just wants speed:

"After my third month filing the same way, TaxPay asked if I wanted it to remember my TaxPro Max login, encrypted, so I could file with one tap. I said yes. Now I just tap 'File' and review before submitting. I can turn it off anytime, and it shows me exactly when it's been used."

14.1 13.1 Final Verdict: Build This

Is it harder than the previous public-first version? Yes. Two bank-data pipelines, a TIN-registration-assist flow, and an optional credential vault are meaningfully more engineering and more risk surface than a single statement-upload pipeline with a strict never-store policy.

Is it better? Yes, for the same reason the public-first pivot itself was better than the B2B-only version: it stops silently excluding people — this time, people without existing bank-credential trust, and people who haven't registered for tax yet — rather than quietly assuming them out of scope.

Can the core of it be built in a hackathon? The MVP — dual bank-data intake, Pidgin/English tax bot, virtual account payment, Guided filing assistant, and a first-time-filer demo — is demo-able in 48-72 hours. Quick Filing Mode is demo-able as an architecture and consent flow, not as a live credential-storage feature; that ships later, deliberately, after review.

The honest truth, unchanged from the last version: the original validation report scored this idea 3/10 for venture-scale potential. That score doesn't change — a Nigerian tax platform won't become a unicorn without FIRS partnership. But it can become a \$1-5M ARR business that tens of thousands of Nigerians use every month, now genuinely open to people the previous version would have turned away at the door. That's not failure. That's a real business, built responsibly.

15 Appendix A: Bank Statement Format Support (MVP)

Bank	PDF (Text)	PDF (Scanned)	CSV	Excel
GTBank	Yes	Yes	Yes	No
UBA	Yes	Yes	Yes	No
Access Bank	Yes	Yes	Yes	No
Zenith Bank	Yes	Yes	Yes	No
First Bank	Yes	OCR fallback	Yes	No
FCMB	Yes	OCR fallback	Yes	No
Sterling Bank	Yes	OCR fallback	Yes	No
Union Bank	Yes	OCR fallback	Yes	No
Fidelity	Yes	OCR fallback	Yes	No
Wema Bank	Yes	OCR fallback	Yes	No
Kuda (Digital)	N/A	N/A	Yes	No
PiggyVest	N/A	N/A	Yes	No
Opay	N/A	N/A	Yes	No
Palmpay	N/A	N/A	Yes	No

Instant Sync (Mono/Okra) coverage, MVP: GTBank, UBA, Access Bank, Zenith Bank, Kuda, Opay confirmed in sandbox; remaining banks fall back to statement upload automatically.

16 Appendix B: Sample Standing Order Instruction Letter

TAXPAY STANDING ORDER INSTRUCTION

Date: [AUTO-GENERATED]

Reference: TAXPAY-SO-[USER-ID]-[MONTH]

To: [BANK NAME]

From: [USER FULL NAME]

Account Number: [USER BANK ACCOUNT]

Please establish a standing order with the following details:

Payee Name: TaxPay Tax Collection
Account Number: [NOMBA VIRTUAL ACCOUNT NUMBER]
Bank: [Nomba Partner Bank]
Amount: ₦[EXACT AMOUNT] ([Amount in words])
Frequency: Monthly
Start Date: [DATE]
End Date: [DATE or "Until further notice"]
Reference: TAXPAY-[USER-ID]-PAYE-[MONTH-YEAR]

This standing order is for monthly tax remittance. The amount may be adjusted periodically – you will receive notification of any changes via SMS from TaxPay.

Signature: _____

Date: _____

For mobile banking setup, use these details:

- Open your [BANK NAME] app
- Go to: Transfers → Standing Order → New
- Enter account number: [VIRTUAL ACCOUNT]
- Enter amount: ₦[AMOUNT]
- Set frequency: Monthly
- Set start date: [DATE]
- Reference: TAXPAY-[USER-ID]-PAYE

17 Appendix C: TaxPro Max Filing Assistant — Auto-Fill Mapping

TaxPay Field	→ TaxPro Max Field Mapping
TIN	→ tax_identification_number
Tax Year	→ assessment_year (auto: current year)
Tax Period	→ return_period (auto: month/year)
Tax Type	→ tax_type (auto: "PAYE")
Gross Tax Liability	→ total_tax_payable
Employee Count	→ number_of_employees
Employee Schedule:	
employee[i].name	→ schedule_row[i].employee_name
employee[i].tin	→ schedule_row[i].employee_tin
employee[i].gross	→ schedule_row[i].gross_income
employee[i].pension	→ schedule_row[i].pension_contribution
employee[i].rent_relief	→ schedule_row[i].consolidated_relief
employee[i].taxable	→ schedule_row[i].taxable_income
employee[i].tax	→ schedule_row[i].tax_due
Computation Sheet:	
total_gross_income	→ comp.gross_total
total_pension	→ comp.total_pension_deduction
total_rent_relief	→ comp.total_consolidated_relief
total_taxable	→ comp.total_taxable_income
total_tax_liability	→ comp.total_tax_payable
Payment (populated only after Nomba webhook confirms payment):	
amount_paid	→ payment.amount
payment_date	→ payment.date
payment_reference	→ payment.reference_number

Note: in both Guided and Quick Filing Mode, the credential field itself is never part of this mapping table – it is either typed directly by the user into TaxPro Max's own form (Guided Mode) or injected transiently by the isolated vault worker at the moment of login (Quick Filing Mode, Section 9) and is never treated as a "TaxPay field" with a stored value in the ordinary sense.

18 Appendix D: Credential Vault — Security Summary (New)

For engineering and review reference, condensed from Section 9:

- **Encryption model:** envelope encryption — per-user Data Encryption Key (DEK) wraps the credential; a KMS-held Key Encryption Key (KEK) wraps the DEK. Application code never handles raw KEK material.
- **Storage:** encrypted credential and wrapped DEK live in an isolated database table, separate from the main user-profile schema, with its own access-control policy.
- **Decryption scope:** only the isolated filing-worker process can request decryption, only for the specific enrolled user, only at the moment a filing is actively initiated.
- **Plaintext lifetime:** exists in worker memory only; never persisted, logged, cached, or included in error reports; explicitly zeroed after the TaxPro Max login step completes.
- **Revocation:** user-triggered, immediate, deletes both the encrypted credential and its wrapped DEK; confirmation receipt issued.
- **Audit trail:** every enroll / use / revoke event logged with actor, timestamp, and outcome; visible to the user via GET /api/v1/vault/access-log.
- **Pre-launch gates:** independent security review of the vault specifically; legal review of FIRS terms-of-service compatibility; NDPR/DPCO scope reassessment. None of these block Guided Mode, which ships without any of this machinery.

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